

HealthPulse AI - AI Tools & Usage Documentation

17 Controlled Capabilities, LangGraph State Engine & Clinical Safety Guardrails

1. LangGraph State Machine Architecture

The AI Agent is constructed using LangGraph (StateGraph). Every conversational turn transitions through structured graph nodes:

- **safety_check_node**: Evaluates incoming transcripts against clinical safety boundaries. If acute symptoms or emergencies are detected, immediately executes transfer_to_human with emergency guidance.
- **intent_recognition_node**: Classifies intent (Discovery, Availability, Booking, Reschedule, Cancellation, Questionnaire, Escalation) and resolves structured context across date shifts.
- **capability_execution_node**: Dynamically executes the required Pydantic-validated capability tool, logging execution timing and parameters.
- **state_synthesis_node**: Generates natural, empathetic voice/chat responses and updates conversation context in persistence.

2. All 17 Controlled Capabilities (Section 3 PRD Compliance)

The AI agent is restricted from making direct SQL or EHR mutations. It must invoke one of the 17 auditable capabilities below:

#	Capability Name	Description & Purpose	Audited Schema
1	search_hospitals	Discovers approved hospitals by city or search query.	city, query
2	search_doctors	Discovers active doctors by specialty, hospital, or name.	specialty, hospital_id
3	check_availability	Calculates real unblocked slots from scheduling service.	doctor_id, date
4	lookup_patient	Retrieves patient demographic record and history.	patient_id, email, phone
5	get_appointment	Fetches booking details scoped to the patient.	appointment_id, patient_id
6	create_appointment	Revalidates slot, creates booking, and verifies in EHR.	hospital_id, doctor_id...
7	reschedule_appointment	Relocates appointment and releases old slot.	appointment_id, new_date...
8	cancel_appointment	Cancels external record and releases availability window.	appointment_id, reason
9	get_questionnaire	Fetches pre-visit form questions for appointment.	hospital_id, appointment_id
10	submit_questionnaire	Saves patient clinical responses and alerts doctor.	appointment_id, responses
11	send_notification	Dispatches in-app, SMS, or email patient alert.	recipient_id, title, message
12	start_workflow	Executes post-confirmation intake workflows.	workflow_type, appointment_id
13	get_context	Retrieves active structured conversation state.	conversation_id
14	update_preferences	Updates patient language and timing preferences.	patient_id, preferences
15	verify_external_appointment	Queries EHR connector to verify booking existence.	external_id, idempotency_key
16	synchronize_state	Reconciles internal state with verified external state.	appointment_id, external_status
17	transfer_to_human	Escalates clinical questions or emergencies to staff.	reason, urgency, department

3. Multi-Turn Context Retention Example

Turn 1 (Discovery): Patient says *"I need an orthopedic doctor this week"*. Agent discovers Dr. Anil Rao (ID: 1) at ABC Hospital (ID: 1) and sets context: {selected_hospital_id: 1, selected_doctor_id: 1}.

Turn 2 (Date Shift): Patient says *"Actually, make that Friday at 10 AM"*. Agent reads context, resolves Friday date, re-validates slot for Dr. Anil Rao without re-asking who the doctor is, and completes booking.